

2.1 Current NSW Electricity Demand Baseline

The current NSW electricity consumption baseline is based on AEMO's aggregated price and demand data for FY2023-24, with significant information summarized in Table 1. NSW's total annual electricity consumption was roughly 65.7 TWh, with a mean half-hourly demand of 7,484 MW and a median of 7,350 MW (AEMO, 2024). The maximum 5-minute interval demand of 13,764 MW was recorded on February 29, 2024, during a summer heatwave, resulting in a wholesale price increase of \$14,811/MWh.

Table 1: NSW Electricity Demand Baseline Statistics, FY2023–24

Statistic	Value
Mean Demand	7,484 MW
Median Demand	7,350 MW
Peak 5-min Interval Demand	13,764 MW (29 Feb 2024, 16:34)
Peak Daily Average Demand	9,893 MW (5 Feb 2024)
Minimum Daily Average Demand	5,870 MW (29 Oct 2023)
Minimum 5-min Interval Demand	3,664 MW
Standard Deviation	1,372 MW
Annual Total Consumption	65,737 GWh (65.7 TWh)
Mean RRP	\$101.57/MWh
Median RRP	\$66.54/MWh

Source: AEMO (2024). Aggregated Price and Demand Data — NSW1, FY2023–24.

NSW demand includes two unique patterns: a seasonal dual-peak driven by winter heating and summer cooling loads, and an intraday duck-curve profile in which noon demand declines due to rooftop solar before skyrocketing at the evening peak between 17:00 and 21:00 (Figures 1-3). Data centers presently consume around 4% of NSW grid-supplied electricity (US Studies Centre, 2026). Unlike residential or commercial

loads, data-center demand is constant and cannot be moved away from high-price peak periods, leaving operators vulnerable to wholesale price volatility.

Figure 1: NSW Daily Average Demand, FY2023–24

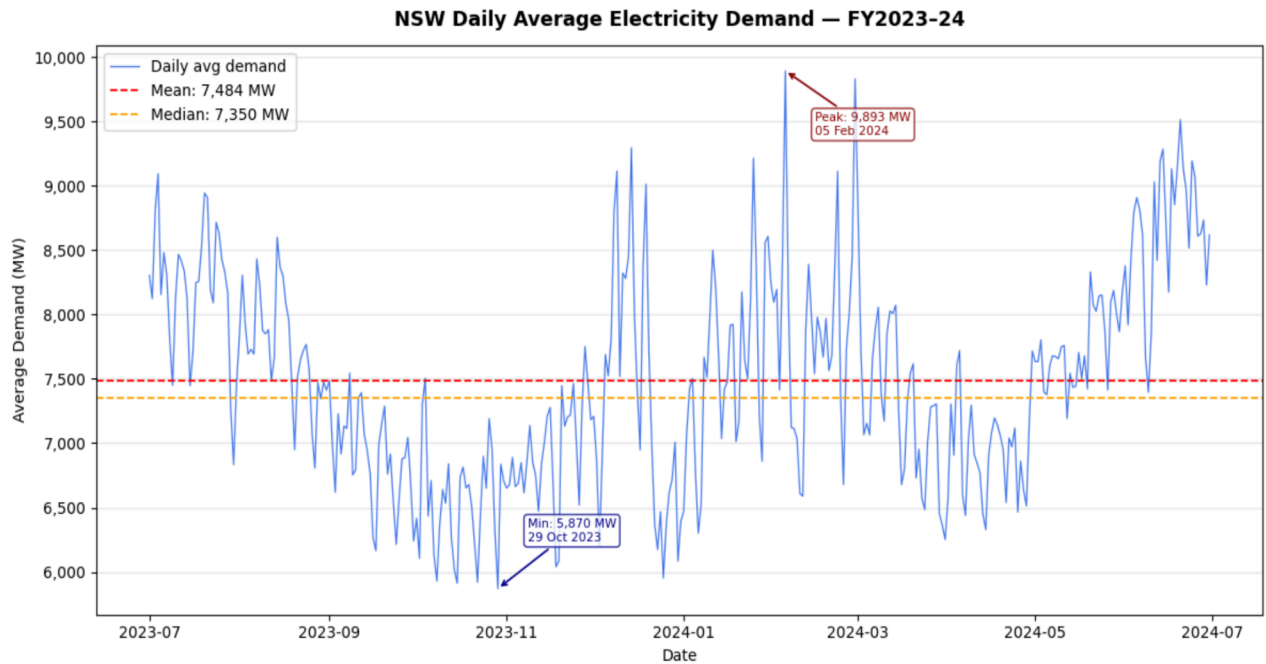


Figure 2: NSW Monthly Electricity Consumption, FY2023–24

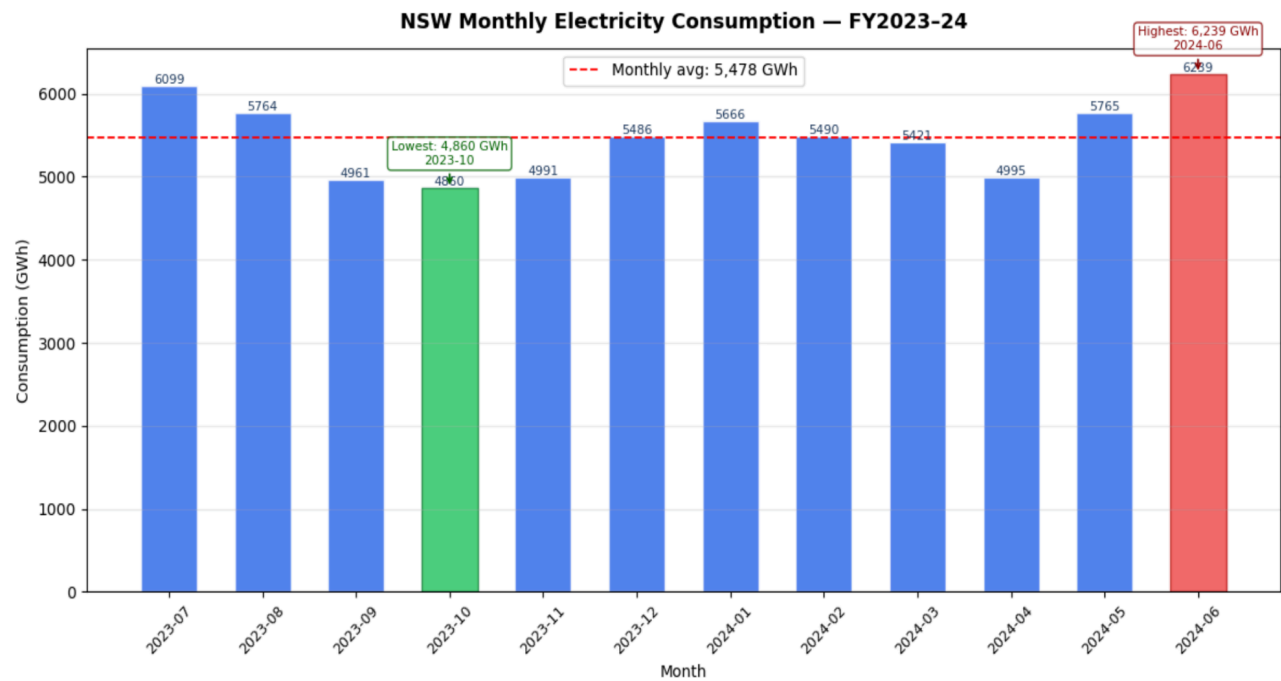
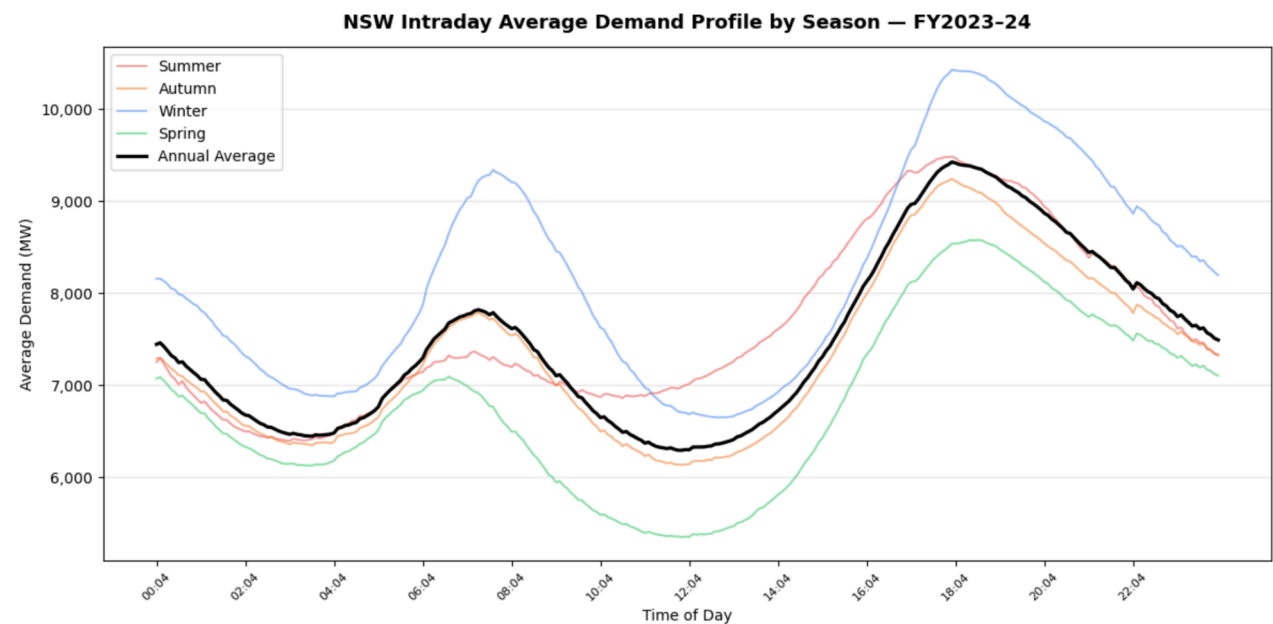


Figure 3: NSW Intraday Average Demand Profile by Season, FY2023–24



This steady, high-intensity load profile is fundamentally different from residential or commercial loads. Unlike other users, data centers cannot shift demand away from high-price peak periods, leaving them particularly vulnerable to wholesale power price fluctuations.

2.2 Data-Center Pipeline and Growth Drivers

The NSW data center market is among the largest and fastest-expanding in Asia-Pacific. Australian data-center electricity consumption has risen by around 15% each year, from 1.4 TWh in FY2018 to 3.9 TWh in FY2025, nearly doubling in seven years (Oxford Economics Australia, 2025). Of this national consumption, NSW is predicted to consume approximately 2.4 TWh, accounting for 60% of national data-center capacity (M3 Property, 2025; author's computation). While NSW accounts for only 33% of total Australian electricity consumption, it contains 60% of national data-center capacity, putting excessive pressure on the NSW grid. Several structural forces are driving this expansion:

- **AI and machine learning workloads:** GPU-based AI training and inference require significantly more energy than traditional cloud computing, resulting in a significant increase in power requirements per rack (AEMO, 2025).
- **Hyperscale investment:** AEMO received 44 GW of data-center connection requests nationwide, indicating unprecedented investment appetite (Oxford Economics Australia, 2025).
- **Growth forecasts:** Under AEMO's Step Change scenario, national data-center consumption is expected to reach 12.0 TWh by FY2030, expanding at a 25.1% annual rate (Oxford Economics Australia, 2025).

Table 2 summarizes key declared NSW projects for 2025-26, focusing on Western Sydney surrounding Eastern Creek, Horsley Park, Marsden Park, and Kemps Creek.

Table 2 — Major Announced NSW Data Center Projects, 2025–26

Operator	Location	Capacity (MW)	Status	Expected Online	Source
NEXTDC S7	Eastern Creek, W Sydney	550 MW	Planning (MOU with OpenAI)	FY2027–28	NEXTDC (2024)
Mamre Road Campus	Kemps Creek, W Sydney	1,200 MW	Under assessment	FY2028–30	Urban Digest (2026)
CDC Data Centers	Marsden Park, NW Sydney	504 MW	Approved	FY2027–28	Urban Developer (2025)
NEXTDC S4	Horsley Park, W Sydney	350 MW	Under construction	Mid-2027	NEXTDC (2024)
AirTrunk SYD3	Huntingwood, W Sydney	320+ MW	Under development	FY2026–27	AirTrunk (2021)
Stockland/EdgeConneX	Kemps Creek, W Sydney	168 MW	Filed (Mar 2026)	FY2027–28	DCD (2026)
Total pipeline		~3,092+ MW			

While the confirmed NSW pipeline capacity is roughly 3,092 MW, this does not transfer directly to grid demand. According to Oxford Economics Australia (2025), 6 out of every 7 MW of connection requests are phantom demand, which stems from speculative applications that are unlikely to materialize on the grid. The scenarios in Section 2.3 are thus based on AEMO's national predictions, scaled to NSW's 60% share, rather than raw pipeline capacity.

2.3 Demand Scenarios: Low, Medium, and High Growth

Three scenarios are presented to address the uncertainties surrounding future NSW data-center demand growth, anchored to AEMO's national predictions from Oxford Economics Australia (2025) and scaled to match NSW's 60% share of national data-center capacity (M3 Property, 2025). The NSW scenario TWh estimations are obtained as follows:

$$\text{NSW TWh} = \text{National AEMO forecast} \times 60\% \text{ NSW share}$$

The implied facility load is back calculated assuming continuous 24/7 operation:

$$\text{Facility MW} = \text{Annual TWh} \times 1,000,000 \div 8,760 \text{ hours}$$

PUE benchmarks reflect the variety of facility types expected in the NSW pipeline: 1.50 for low-growth (older colocation facilities), 1.40 for medium-growth (modern facilities), and 1.30 for high-growth (new hyperscale builds with best-practice efficiency) (Uptime Institute, 2024; Socomec, 2025; AKCP, 2024). Utilization rates of 55%, 65%, and 75% correspond to typical hyperscale ramp-up trajectories. The three possibilities are summarized in Table 3.

Table 3 — NSW Data-Center Demand Growth Scenarios, FY2025–FY2034

Parameter	Low Growth	Medium Growth	High Growth
AEMO scenario reference	Progressive Change	Step Change	Accelerated DC Growth
National DC forecast (FY2030)	8.5 TWh	12.0 TWh	~17.0 TWh
NSW share of national capacity	60%	60%	60%
NSW new additions by FY2034	5.1 TWh	7.2 TWh	10.2 TWh
Average PUE	1.50	1.40	1.30
Average utilization rate	55%	65%	75%
Implied facility load (MW)	582 MW	822 MW	1,164 MW
NSW additions as % of baseline	7.8%	11.0%	15.5%
Total NSW DC by FY2034	7.5 TWh	9.6 TWh	12.6 TWh
Total DC as % of baseline	11.4%	14.6%	19.2%

Note: NSW's new additions derived by scaling national DC forecasts by NSW's 60% share of national data-center capacity. Sources: Oxford Economics Australia (2025); AEMO (2025); M3 Property (2025); Uptime Institute (2024); Socomec (2025); AKCP (2024)

The low-growth scenario (7.5 TWh, 11.4% of the NSW baseline) includes just projects that are currently under construction, which is consistent with M3 Property's (2025) assessment that only 25% of the pipeline is development-ready. The medium-growth scenario (9.6 TWh, 14.6%) is the base case, which is precisely congruent with the US Studies Centre's (2026) forecast that NSW data centers will consume around 11% of state electricity by FY2030. The high-growth scenario (12.6 TWh, 19.2%) assumes near-full pipeline realization under AEMO's Accelerated DC Growth Strategy, a structural shift that necessitates major investment in generation and transmission infrastructure.

Figure 4 — NSW New Data-Center Demand Additions by Scenario, FY2025–FY2034

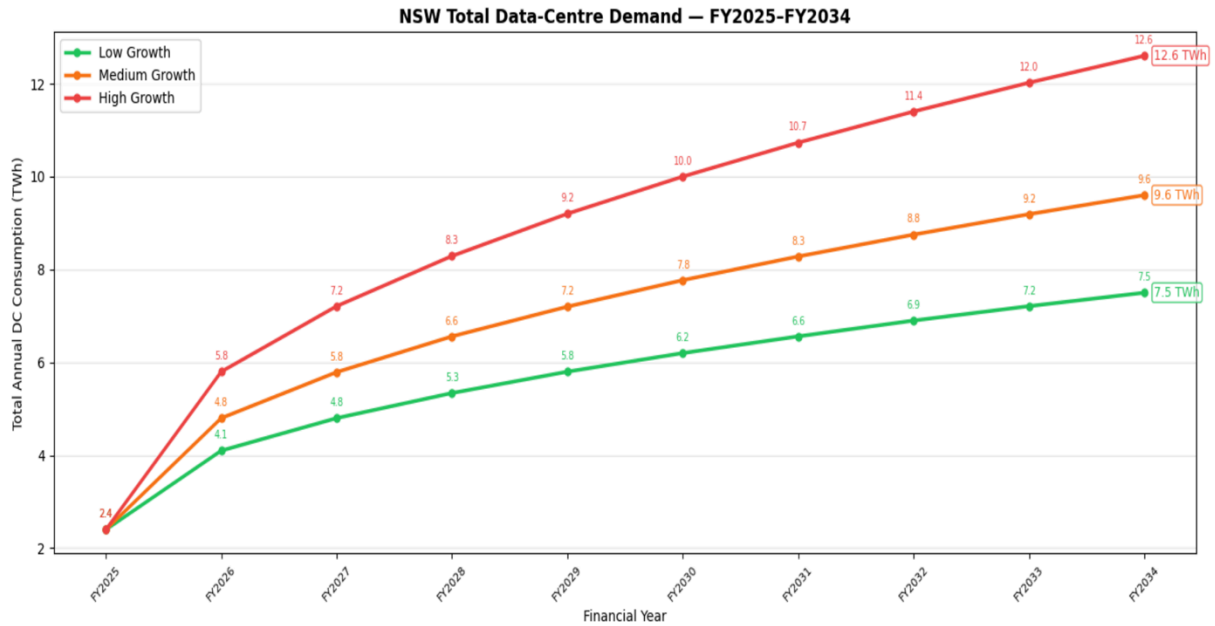


Figure 5 — NSW New Data-Center Demand as % of NSW Baseline, FY2025–FY2034

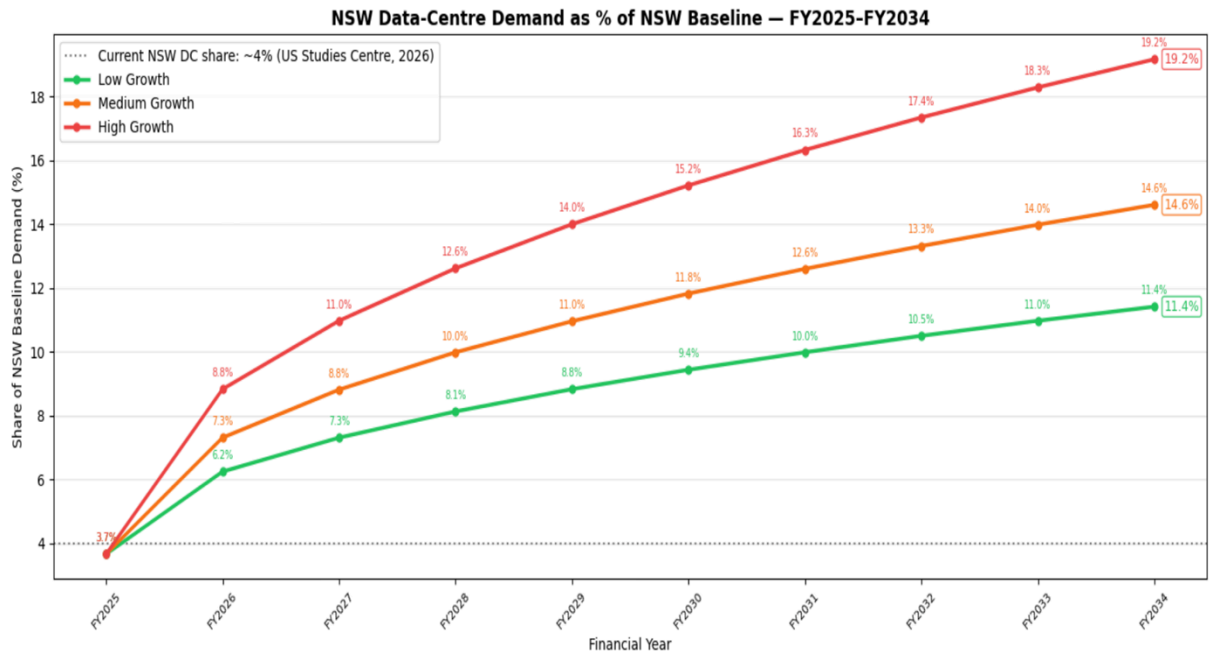
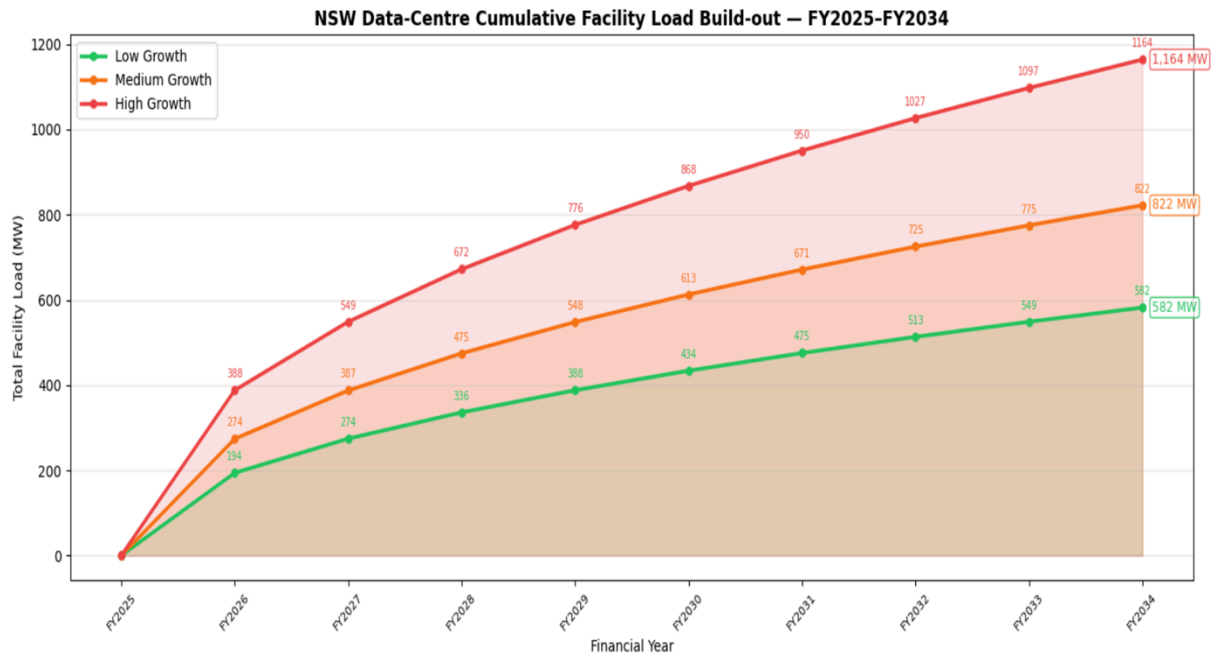


Figure 6 — NSW Data-Center Cumulative Facility Load Build-out, FY2025–FY2034



2.4 Significance Relative to NSW Total Demand

Table 4 and Figures 7.1 and 7.2 explain the scenario estimates by comparing total NSW data center demand to the NSW electricity baseline. NSW data centers presently account for around 4% of grid-supplied electricity (US Studies Centre, 2026), and this percentage is expected to increase significantly by FY2034 under all three scenarios.

Table 4.1 — Total NSW DC Demand vs Static Baseline (65.7 TWh)

<i>Year</i>	<i>NSW Baseline</i>	<i>Low Growth</i>	<i>Medium Growth</i>	<i>High Growth</i>
FY2025	65.7 TWh	2.4 TWh (3.7%)	2.4 TWh (3.7%)	2.4 TWh (3.7%)
FY2026	65.7 TWh	4.1 TWh (6.2%)	4.8 TWh (7.3%)	5.8 TWh (8.8%)
FY2027	65.7 TWh	4.8 TWh (7.3%)	5.8 TWh (8.8%)	7.2 TWh (11.0%)
FY2028	65.7 TWh	5.3 TWh (8.1%)	6.6 TWh (10.0%)	8.3 TWh (12.6%)
FY2029	65.7 TWh	5.8 TWh (8.8%)	7.2 TWh (11.0%)	9.2 TWh (14.0%)
FY2030	65.7 TWh	6.2 TWh (9.4%)	7.8 TWh (11.9%)	10.0 TWh (15.2%)
FY2031	65.7 TWh	6.6 TWh (10.0%)	8.3 TWh (12.6%)	10.7 TWh (16.3%)
FY2032	65.7 TWh	6.9 TWh (10.5%)	8.8 TWh (13.4%)	11.4 TWh (17.4%)
FY2033	65.7 TWh	7.2 TWh (11.0%)	9.2 TWh (14.0%)	12.0 TWh (18.3%)
FY2034	65.7 TWh	7.5 TWh (11.4%)	9.6 TWh (14.6%)	12.6 TWh (19.2%)

*Note: NSW baseline held constant at FY2023–24 levels. Percentages represent total DC demand as % of 65.7 TWh.
Sources: AEMO (2024); Oxford Economics Australia (2025); M3 Property (2025).*

Table 4.2 — Total NSW DC Demand vs Growing Baseline (NEM CAGR 2.55% p.a.)

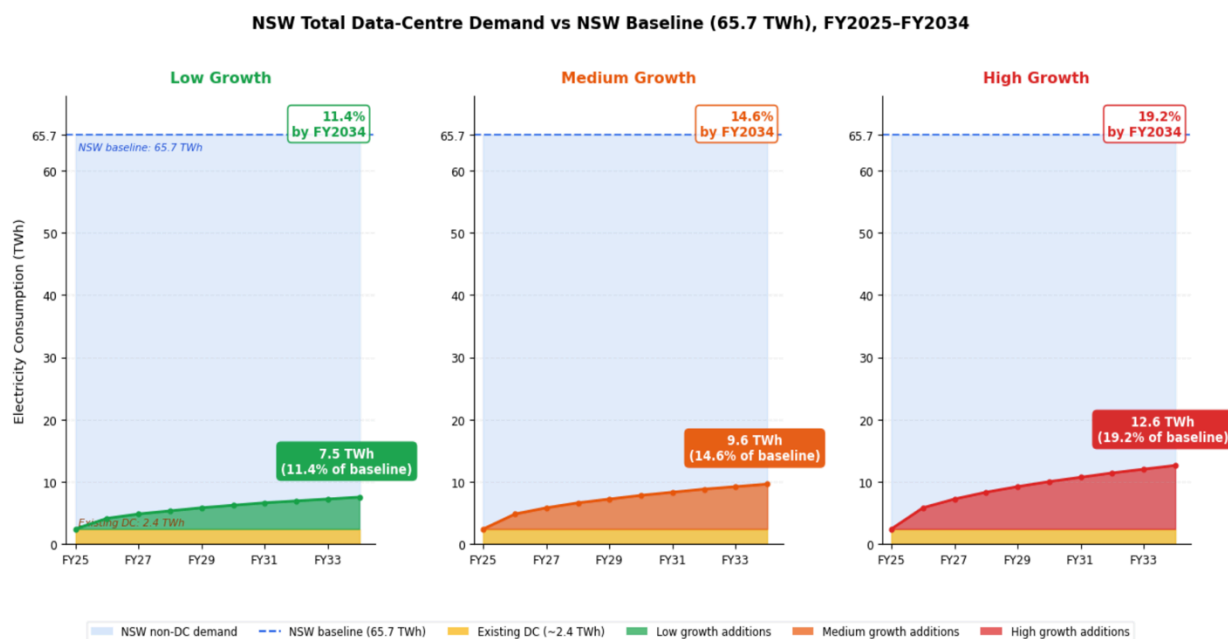
Year	NSW Projected Baseline	Low Growth	Medium Growth	High Growth
FY2025	65.7 TWh	2.4 TWh (3.7%)	2.4 TWh (3.7%)	2.4 TWh (3.7%)
FY2026	67.4 TWh	4.1 TWh (6.1%)	4.8 TWh (7.1%)	5.8 TWh (8.6%)
FY2027	69.1 TWh	4.8 TWh (6.9%)	5.8 TWh (8.4%)	7.2 TWh (10.4%)
FY2028	70.9 TWh	5.3 TWh (7.5%)	6.6 TWh (9.3%)	8.3 TWh (11.7%)
FY2029	72.7 TWh	5.8 TWh (8.0%)	7.2 TWh (9.9%)	9.2 TWh (12.7%)
FY2030	74.5 TWh	6.2 TWh (8.3%)	7.8 TWh (10.5%)	10.0 TWh (13.4%)
FY2031	76.4 TWh	6.6 TWh (8.6%)	8.3 TWh (10.9%)	10.7 TWh (14.0%)
FY2032	78.4 TWh	6.9 TWh (8.8%)	8.8 TWh (11.2%)	11.4 TWh (14.5%)
FY2033	80.4 TWh	7.2 TWh (9.0%)	9.2 TWh (11.4%)	12.0 TWh (14.9%)
FY2034	82.4 TWh	7.5 TWh (9.1%)	9.6 TWh (11.7%)	12.6 TWh (15.3%)

Note: NSW baseline projected using NEM-wide CAGR of 2.55% p.a. (AEMO, 2025 ESOO). NSW-specific growth rate unavailable, NEM rate used as proxy. Sources: AEMO (2025); Oxford Economics Australia (2025); M3 Property (2025).

Figure 7.1 assumes a static baseline of 65.7 TWh as an upper-bound estimate of the data-center share. Figure 7.2 uses an increasing baseline with a NEM-wide CAGR of 2.55% per year (AEMO, 2025), reducing predicted shares to 9.1% (low growth) to 15.3% (high growth) by FY2034. The medium scenario for FY2030, 11.9% below the static baseline, is explicitly supported by the US Studies Centre's (2026) prediction of about 11% by FY2030.

Peak-demand effective facility loads of 582 MW, 822 MW, and 1,164 MW account for 7.8%, 11.0%, and 15.6% of NSW's current mean demand of 7,484 MW (AEMO, 2024). Since data-center demand is consistent and inflexible, these expansions put permanent upward pressure on system stress, especially during evening peak periods when prices are especially vulnerable to increased load.

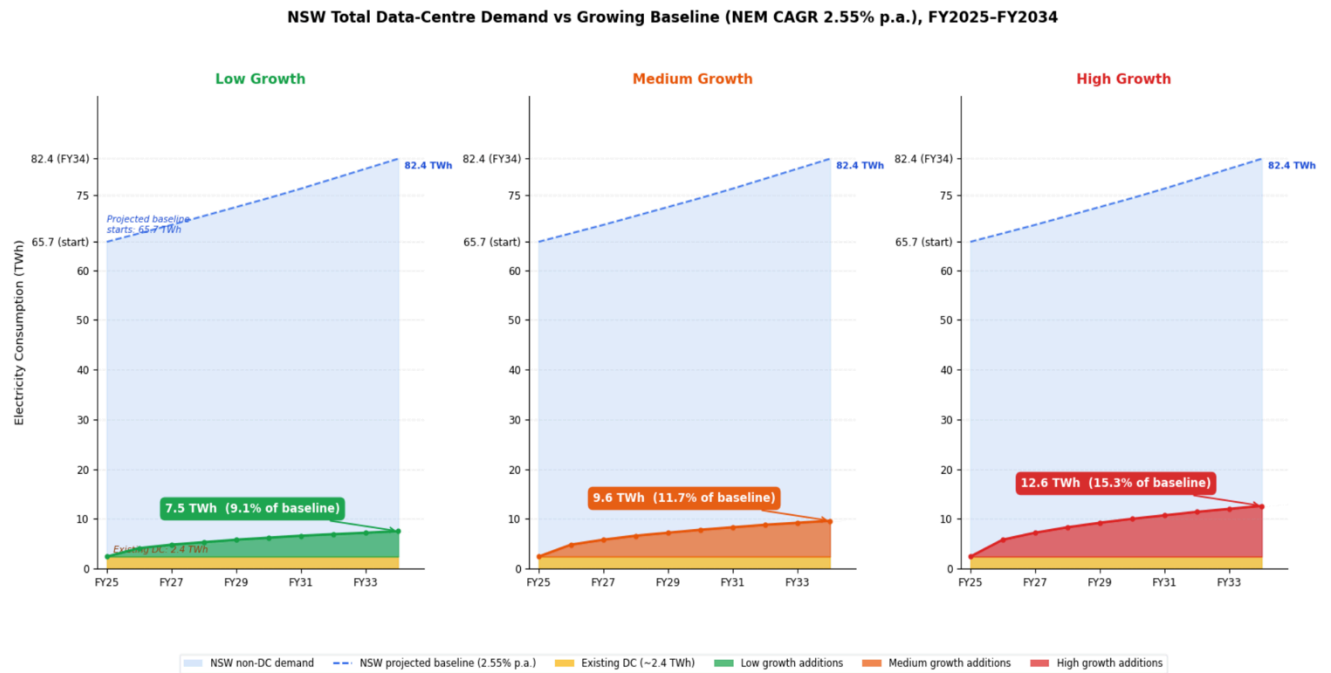
Figure 7.1 — NSW Total Data-Center Demand vs Static Baseline, FY2025–FY2034



Note: NSW baseline held constant at FY2023–24 levels. Percentages represent total DC demand as % of 65.7 TWh.

Sources: AEMO (2024); Oxford Economics Australia (2025); M3 Property (2025).

Figure 7.2 — NSW Total Data-Center Demand vs Growing Baseline, FY2025–FY2034

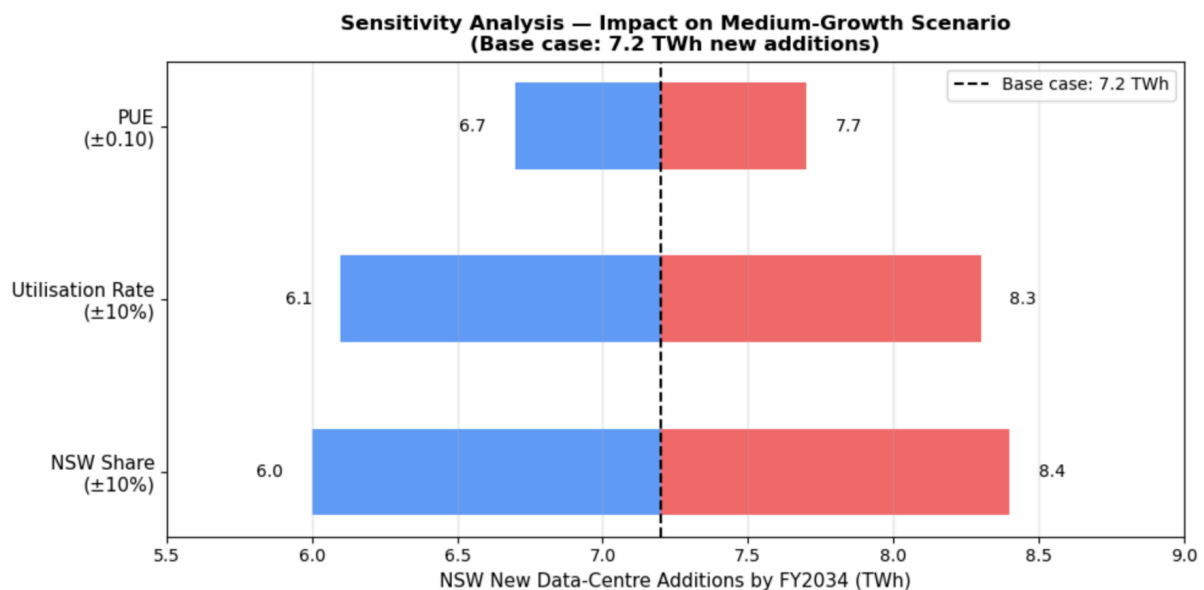


Note: NSW baseline projected using NEM-wide CAGR of 2.55% p.a. (AEMO, 2025 ESOO). NSW-specific growth rate unavailable, NEM rate used as proxy. Sources: AEMO (2025); Oxford Economics Australia (2025); M3 Property (2025).

2.5 Sensitivity Analysis and Key Uncertainties

The sensitivity analysis focuses exclusively on new data center installations, using the medium-growth base assumption of 7.2 TWh, with the current NSW baseline of around 2.4 TWh assumed as fixed. Three fundamental assumptions are evaluated while the others remain constant (Figure 8).

Figure 8 — Sensitivity Analysis Tornado Chart



The NSW share of national capacity is the most sensitive assumption, with a $\pm 10\%$ variance resulting in a range of 6.0 - 8.4 TWh. This reflects uncertainty surrounding M3 Property's (2025) projection of Sydney hosting 60% of national capacity. The second most relevant factor is utilization rate, which ranges between 55% and 75% and results in a range of 6.1 - 8.3 TWh, driven by uncertainties about AI workload adoption. PUE has the least impact (1.0 TWh) since it affects the split between IT equipment and cooling overhead rather than total grid demand. All estimates should be viewed as directional rather than accurate and updated as recent AEMO forecasts and market data become available (Oxford Economics Australia, 2025).

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